

BI Assessment: Implementing a Predictive Model for Video Views in Specific Point in the Future



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Project Overview

The goal of this task is to develop a machine learning model capable of predicting the number of views a video will receive at a specific point in the future. The model will progressively improve its predictions by taking additional observations as input. For example, the model will first make predictions based on a single observation, then refine its predictions with two observations, and so on. This approach leverages the temporal nature of the data, allowing the model to incorporate trends and patterns over time.

Key Objectives

1. **Design a flexible model architecture** to accommodate varying numbers of observations as input.
2. **Preprocess and prepare the dataset** for training, ensuring it is optimized for temporal predictions.
3. **Train and validate the model** to accurately predict video views at future time points.
4. **Evaluate model performance** using appropriate metrics and document the results.

Dataset Overview

The dataset contains various features for videos, including scrape timestamps and future view counts for specific moments in time. This temporal structure provides a rich foundation for predictive modeling.

Download: [Dataset](#)

Column Descriptions

Column	Type	Description	Notes
VIDEO_ID	Categorical	Unique identifier for each video.	Dropped during preprocessing.
TITLE	Categorical	Video title.	Dropped due to high uniqueness.
DESCRIPTION	Categorical	Video description text.	Not used due to missing values.
VIDEO_DURATION	Numerical	Duration of the video in seconds.	Useful for viewer engagement trends.
VIDEO_PUBLISHED_AT	Timestamp	Date and time of video publication.	Dropped for baseline model.
VPH	Numerical	Current views per hour.	Significant for traction analysis.
LIKES	Numerical	Number of likes received.	Strongly correlated with views.
VIEWS	Numerical	Total current view count for each video.	Log-transformed to reduce skewness.
COMMENTS	Numerical	Number of comments.	Indicates engagement.
CHANNEL_SUBSCRIBERS	Numerical	Subscriber count for the hosting channel.	Reflects channel popularity.
VIEWS_IN_FUTURE_TIME	Numerical	Future view count at a specific time.	Target variable.
TIME_IDX	Numerical	Time index relative to each	Essential for temporal

		video's scrape timestamp.	modeling.
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Example Data

VIDEO_ID	TITLE	DESCRIPTION	VPH	LIKES	VIEWS	COMMENTS	CHANNEL_SUBSCRIBERS	SCRAPE_TIME	VIEWS_IN_FUTURE_TIME	TIME_IDX
-3bV2e-NKE	A Girl's Ego	My Hero Academia...	823	374	2477	64	901,000	2024-10-08 22:20:26.467	10,023	0
-3bV2e-NKE	A Girl's Ego	My Hero Academia...	976	509	3386	92	901,010	2024-10-08 22:30:25.600	10,023	1

Deliverables

1. **Preprocessed Dataset:** Cleaned and feature-engineered data.
2. **Trained Model:** A Python implementation that can take varying numbers of observations.
3. **Evaluation Report:** Summary of model performance.
4. **Code Repository:** Well-documented Python scripts.

Contact

giannis.prokopiou@orfium.com

eleftheria@orfium.com